



Assessing the regional digital divide across the European Union-27

María Rosalía Vicente*, Ana Jesús López

Applied Economics, University of Oviedo, Campus del Cristo s/n, Oviedo-Asturias 33006, Spain

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ABSTRACT

Despite the efforts devoted to the analysis of the digital divide over the last years, the regional evidence on this issue is still scarce. Within this context, the goals of this paper are twofold: (1) measuring the digital divide across the regions of the 27 Member States and within each country and (2) explaining the observed regional disparities. The analysis leads to identify Dutch regions as the Top-10 in ICT, while Greece and Bulgaria occupy the Bottom-10. Therefore, results show that the regional digital divide reflects to some extent the income gap. However, regional policy seems to be having some positive implications for technology adoption. In particular, the rural-versus-urban dimension of the digital gap appears to be less important than it is usually claimed to. Likewise, some evidence is found of the role of cultural and institutional factors in ICT adoption, an issue that has not been previously analyzed at the regional level.

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1. Introduction

The last two decades have seen the revolutionary spread of Information and Communication Technologies (ICT). The developments in hardware, software, and electronics have lead to affordable general purpose technologies (Bresnahan & Trajtenberg, 1995) that can be used in most sectors and whose benefits may extend everywhere. Thus, e-mail, web-surfing, online job-search, and social networks have become integrated in daily life. Likewise, activities such as e-learning, e-government, and e-commerce are more and more a major element in the life of many individuals.

Hence, ICT have been recognized as a major contributor in social and economic development (OECD, 2004). Furthermore, several authors have stressed the importance of these technologies as a catalyst for growth in the current economic crisis (WEF, 2009). In fact, both American and European recovery plans have considered ICT as part of their strategic actions: for instance, the European Commission has earmarked EUR 1 billion of extra spending for investment in broadband, with special attention to the improvement of high-speed connections in rural regions (European Commission, 2009).

ICT are increasingly determining the ability of individuals, firms, and territories to remain competitive and to do things in a more effective and efficient way. Therefore, the inequalities detected in their diffusion may have serious implications for economic growth, human development, and the creation of wealth (ITU, 2006). For this reason, the so-called digital divide has become a major issue on both international and national forums (ITU & UNCTAD, 2007; World Bank, 2009).

According to the OECD (2001) “the term digital divide refers to the gap between individuals, households, businesses and geographic areas at different socio-economic levels with regard both to their opportunities to access information and communication technologies and to their use of the Internet for a wide variety of activities”.

* Corresponding author. Tel.: +34 985 10 50 53; fax: +34 985 10 5050.

E-mail addresses: mrosalia@uniovi.es (M.R. Vicente), anaj@uniovi.es (A.J. López).

The concept of the digital divide is complex, with several layers and dimensions that may overlap (OECD, 2009). Hence, its analysis requires taking into account a large range of technologies, variables, and geographical settings (Billón, Marco, & Lera-López, 2009b; OECD, 2009).

In this context, two main approaches to the digital divide have emerged in the literature: one focuses on measuring the divides, and the other tries to explain the determinants of such gaps (Corrocher & Ordanini, 2002). Analytical work has been carried out in both areas by means of several methodologies and statistical techniques (OECD, 2009). In the first approach, a common practice has been to build composite indicators in order to reveal the relative positions (of countries, usually) in ICT adoption (Dutta & Jain, 2004; ITU, 2009; Orbicom, 2002; Selhofer & Hüsing, 2002; Wolcott, Press, McHenry, Goodman, & Foster, 2001). Other authors (Çilan, Bolat, & Coşkun, 2009; Corrocher & Ordanini, 2002; Vicente & López, 2006a), however, have preferred multivariate methods to assess the complexity of the digital divide. In the second approach, multiple regression analysis has been mostly used in order to explain the number of Internet users (Beilock & Dimitrova, 2003; Chinn & Fairlie, 2007, 2010; Guillén & Suárez, 2001, 2005; Tanner, 2003), together with hosts (Hargittai, 1999; Kiiski & Pohjola, 2002; Oyelaran-Oyeyinka & Lal, 2005), and mobile telephones (Ahn & Lee, 1999; Bagchi, 2005; Dasgupta, Lall, & Wheeler, 2005; Quibria, Shamsun, Tschanh, & Reyes-Macasaquit, 2003; Wong, 2002), among other variables. Nonetheless, it is difficult to properly evaluate the digital progress of a territory by focusing on just one technology (Wolcott et al., 2001). For this reason, Billón, Marco, & Lera-López (2009b) use canonical correlation analysis in order to take together a large number of dependent variables (computers, Internet users, broadband subscribers, mobile phones, telephones mainlines, and secure Internet servers).

Attempts to combine the two approaches to the digital divide, measuring and explaining it, are few. Bagchi (2005) is an exception and builds a composite indicator on four variables that characterize digital development in the OECD and Latin American countries, and then explains the obtained index by means of regression analysis.

All these previous studies have generally focused on countries as the relevant geographical setting for the analysis of the digital divide. Research at the regional level is much scarcer and mostly refers to the United States, on both the papers measuring the regional digital gap (Atkinson & Andes, 2008; Progressive Policy Institute, 2002) and those explaining regional Internet diffusion (Chaudhuri, Flamm, & Horrigan, 2005; Grubestic, 2006; Horrigan, Stolp, and Wilson, 2006; Kolko, 2000; Mills & Whitacre, 2003; US Department of Commerce, 2000).

References to the European Union (EU) are even fewer because of the limited access to comparable regional statistics. Nonetheless, it is worth mentioning that some European projects tried to remedy this deficiency (BISER, 2004; UNDERSTAND, 2006). As a consequence, most research has focused either on the measurement of regional ICT disparities within a particular member state (Vicente & López, 2007), or on explaining a single dependent variable such as Internet users (Billón, Ezcurra, & Lera-López, 2008), or enterprises with a website (Billón, Ezcurra, & Lera-López, 2009a).

In this context, this paper contributes to the existing literature in four main ways. In the first place, the two approaches to the digital divide are joined; hence, not only the digital gap is measured but also explained. Though Bagchi (2005) has done this previously, the measurement carried out in this paper is based on a larger number of Internet-related variables.¹ These variables are then combined into an index by means of factor analysis, a multivariate technique which has been previously used successfully in summarizing the multiple dimensions of the digital divide. Furthermore, the gap is measured at the regional level, while Bagchi (2005) carries out a cross-country analysis. The factor-derived index allows to build a ranking of regions and identify best-and-worst performing European territories in ICT, this being the first attempt to do it in the literature. In the second place, the methodological approach employed to in this paper allows to quantify the global digital gap not only across the full European Union, but also within each member state. In the third place, this paper covers, for the year 2008, 164 regions across the 27 member states of the European Union (EU-27), extending the area considered in previous analyses.² Finally, some cultural and institutional-related elements are included in the regression model in order to explain ICT scores. Although this kind of determinants has been considered for explaining Internet adoption across countries, this is the first time it is included in a regional model.

The paper is structured as follows. The next section presents a literature review on the digital divide and gives the conceptual framework for the analysis (Section 2). Then, the focus is on the measurement of the regional digital divide, describing the dataset, methodology and results in the form of a regional ranking (Section 3). In order to get further insights over this issue, the determinants of the regional position in ICT are investigated (Section 4). Finally, some concluding remarks are discussed (Section 5).

2. Literature review

In the literature of the digital divide, two main types of studies can be distinguished: one concentrated on measuring and quantifying the extent, evolution, and pace of the digital divide (Corrocher & Ordanini, 2002; OECD, 2009; Orbicom, 2002),

¹ Bagchi (2005) develops an index based on four technologies (telephones, computers, cell phones, and Internet use), while this paper focuses more on Internet use and online services.

² In the BISER project, Milicevic and Gareis (2003) take into account 28 European regions. The UNDERSTAND project focuses on the analysis of nine different regions across the fifteen old member states of the Union (EU-15). Demunter (2005) makes a single distinction between Objective 1 regions and others. Billón et al. (2008) cover 76 regions in EU-15. Although Billón et al. (2009a) consider 239 regions of the enlarged Union (a number fairly higher than the one analyzed in this paper), these authors do not cover the full EU since some countries (such as Cyprus, Denmark, Luxembourg, Malta, Lithuania, Latvia, Estonia, and Slovenia) had to be excluded from the analysis due to the lack of data. Moreover, their analysis refers to the year 2002.

and other focused on explaining the determinants of ICT disparities (Andonova, 2006; Billón et al., 2009b; Chinn & Fairlie, 2007, 2010; Demoussis & Giannakopoulos, 2006; Hargittai, 1999; Kiiski & Pohjola, 2002; Quibria et al., 2003).

Analytical work has been conducted in both areas, but from different methodological approaches and using several statistical techniques, depending on each of these goals (measuring and quantifying the digital gap, or explaining it) (OECD, 2009).

Both approaches have mostly focused on the analysis of the cross-country digital divide (Chinn & Fairlie, 2007, 2010; Hargittai, 1999; Oyeleran-Oyeyinka & Lal, 2005), while research at the regional level is much scarcer due mainly to data limitations. However, there are some exceptions (on measuring the regional digital divide see Atkinson & Andes, 2008; Milicevic & Gareis, 2003; Progressive Policy Institute, 2002; and Vicente & López, 2007; and on explaining its determinants see Billón et al., 2008, 2009a; Grubestic, 2006; and Kolko, 2000, among others). In a related line of research, the explanatory factors of the urban–rural digital gap have also been investigated by means of microdata (Chaudhuri et al., 2005; Mills & Whitacre, 2003; Noce & McKeown, 2008; US Department of Commerce, 2000). Furthermore, some authors have considered not only individuals' characteristics but also those of the region/county where he/she lives in order to explain the digital divide (Demoussis & Giannakopoulos, 2006; Horrigan et al., 2006; Schleife, 2010).

Among the studies measuring the digital divide, there has been an increasing number of attempts to quantify the divide and catch its multidimensionality by means of composite indicators or indices (Al-Mutawakil, Heshmati, & Hwang 2009; Bagchi, 2005; Corrocher & Ordanini, 2002; Dutta & Jain, 2004; Economist Intelligence Unit, 2009; ITU, 2009; Orbicom, 2002; Selhofer & Hüsing, 2002; Wolcott et al., 2001).³ These indices have revealed as useful tools for summarizing the complexity of the digital divide across territories, and for easily interpreting the gaps by building rankings. For instance, Atkinson and Andes (2008) develop an index including 29 variables in order to rank each American state on the extent to which its economy is progressing towards a digital one and is ready to compete nationally and globally.

In this context, several authors have successfully incorporated the various dimensions of digital development into an index by means of factor analysis (Al-Mutawakil et al., 2009; Corrocher & Ordanini, 2002; ITU, 2009; Soupizet, 2004). Using this technique, Corrocher and Ordanini (2002) identify six factors that then aggregate in a synthetic index in order to measure the digital divide across ten developed countries. The ITU (2009) creates the ICT development index (IDI) including eleven indicators associated to three ICT-related dimensions (infrastructure and access, use, and skills). In this case, factor analysis is used (1) to check whether the considered dimensions are statistically well-balanced, (2) to investigate the association between the indicators and how they change in relation to each other, and (3) to identify the most relevant indicators to incorporate in each of the dimensions. Then, the IDI is used to measure the global digital divide and examine how it has developed in recent years. Vicente and López (2006a, 2007) combine factor analysis with cluster techniques over a set of ICT-related indicators in order to classify the levels of digital development across European countries and Spanish regions, respectively.

Some other multivariate techniques have also been used to measure the digital divide. For instance, Çilan et al. (2009) employ both multivariate analysis of variance and discriminant analysis to explore whether a digital divide exists among the old, the new, and the candidate member states of the European Union.

Studies aimed at explaining ICT adoption have mostly used multiple regression analysis, with Internet users being the most often considered dependent variable (Andrés, Cuberes, Mame, & Serebrisky, 2010; Beilock & Dimitrova, 2003; Chinn & Fairlie, 2007, 2010; Guillén & Suárez, 2001, 2005; Tanner, 2003) together with hosts (Hargittai, 1999; Kiiski & Pohjola, 2002; Oyeleran-Oyeyinka & Lal, 2005), and mobile telephones (Ahn & Lee, 1999; Bagchi, 2005; Dasgupta et al., 2005; Quibria et al., 2003; Wong, 2002). Other authors have focused on secure e-commerce hosts (Gibbs & Kraemer, 2004; Wong, 2002), expenditure on computer hardware per capita (Pohjola, 2003), imports of computer equipment (Caselli & Coleman, 2001), and most recently broadband penetration (Cava-Ferreruela & Alabau-Muñoz, 2006; Distaso, Lupi, & Manenti, 2006).

In comparison to these analyses of single dependent variables, Billón et al. (2009b) use canonical correlation analysis in order to take together a large number of dependent variables (PC, Internet users, broadband subscribers, mobile phones, telephones mainlines, and secure Internet servers).

Empirical research shows that the main factor underlying the disparities in ICT adoption is differences in economic wealth, both between territories and between individuals. (Andonova, 2006; Chinn & Fairlie, 2007, 2010; Crenshaw & Robison, 2006; DiMaggio, Hargittai, Celeste, & Shafer, 2004; Fairlie, 2004; Hargittai, 1999; ITU, 2006; Kiiski & Pohjola, 2002; Kraemer, Ganley, & Dewan 2005; Norris, 2001; Pohjola, 2003). In fact, Chinn and Fairlie (2007, 2010) find that economic wealth, represented by income per capita, is the biggest single factor explaining the disparities in computer and Internet penetration rates. More concretely, 43.7 and 20.6 percentage points of the gaps in computer and Internet penetration, respectively, are due to the existing income differences between developed and developing countries (Chinn & Fairlie, 2010).

Income usually shapes the patterns of ICT diffusion because it is associated with some critical factors for ICT supply and demand, being the interaction of these forces what defines the diffusion process (Karshenas & Stoneman, 1995).

On the one hand, income is related to telecommunications infrastructure. The adoption of a new technology may depend on the availability of specialized infrastructure; in such cases, better-equipped areas will be among the first to adopt (Billón et al., 2009b). Therefore, the higher income of a territory, the larger availability of resources to invest in the required

³ See Vicente and López (2008) for complete survey on the development of ICT indices.

telecommunications infrastructure (Fink & Kenny, 2003), and consequently, the higher the rate of ICT diffusion (Billón et al., 2008; Chinn & Fairlie, 2007; Corrocher & Ordanini, 2002; ITU, 2009; Quibria et al., 2003).

On the other hand, income is linked to factors such as human capital and socio-demographic features of the population, which are crucial to explain ICT demand (Billón, Lera, & Marco, 2010; Chinn & Fairlie, 2010; Fairlie, Beltran, & Das, 2010; Hargittai, 1999; Kiiski & Pohjola, 2002; Kraemer et al., 2005; Quibria et al., 2003; Tanner, 2003).

Most studies point out that the adoption of innovations is faster among those with high education levels (Rogers, 1983). In the case of the Internet, which is *par excellence* an interactive technology, any user needs the access to infrastructure as well as the appropriate skills to search for information, to critically review it, and use it (Vicente & López, 2006b). This implies that those territories with better educated populations will be more prone to exhibit higher levels of ICT penetration than those with less skilled individuals (Crenshaw & Robison, 2006; Hargittai, 1999; Vicente & López, 2006b). Billón et al. (2008, 2009a) find that the regional proportion of population with tertiary studies is positively correlated with Internet and e-commerce adoption. In addition, Norris (2001) indicates that academic institutions have been important drivers of Internet diffusion as they were among the first institutions that decided to go online. In this same sense, Goldfarb (2006) finds that those individuals, who learned how to use the Internet at university in the mid-1990s, kept as users after graduation, a fact that underlines “a micro-level role for academic institutions in diffusion”.

Population size and its socio-demographic features (for instance, age, racial group, density, and urbanization) also explain the digital divide across territories (Bagchi, 2005; Chinn & Fairlie, 2007, 2010; Fairlie, 2004; Kraemer et al., 2005; Quibria et al., 2003).

Empirical research based on microdata has largely shown that ICT adoption is mainly associated to the lives of the younger generations (Fairlie, 2004; OECD, 2001; US Department of Commerce, 2000). The age gap in ICT use might be the result of a range of factors including skills and attitudes, as well as perceived needs (Cohendet, 2003; Sciadas, 2002). Children may as well generate a stronger demand for technology, especially for educational purposes (Fairlie et al., 2010). In this context, it is expected that territories with a high proportion of young population (old population) will have high (low) Internet penetration rates (Chinn & Fairlie, 2007).

Furthermore, several studies have investigated the determinants of the digital divide between racial groups (Fairlie, 2004; Novak & Hoffman, 1998; US Department of Commerce, 2000). Such gaps appear to be mainly related, but not exclusively, to income and education inequalities (Fairlie, 2004). In this line, Schleife (2010) finds a negative association between the proportion of foreigners and the local rate of Internet users due possibly to linguistic/communication problems or educational deficiencies of the former group.

Population size, density and the degree of urbanization are also correlated with the digital divide. The urban density theory highlights the negative association between ICT adoption costs and population size and density (Forman, Goldfarb, & Greenstein 2005). The access to these technologies is easier and cheaper in cities (than in rural areas) because they have better telecommunications infrastructure, and the costs of the deployment of new infrastructure are lower. Besides, cities tend to concentrate high-skilled labor and knowledge resources (Baptista, 2001; Kaufmann, Lehner, & Todtling, 2003; Zitt, Barre, Sigogneau, & Lavilla, 1999). Likewise, faster network connections are available in highly populated areas compared to low-density territories. In this sense and from the perspective of social learning, people living in cities may have lower costs to start using the Internet, because they can learn how to use it from their neighbors (Agarwal, Animesh, & Prasad, 2009). Given that the Internet spreads as early adopters share their experiences and knowledge with other individuals (Rogers, 1983), spatial proximity positively influences diffusion (Billón et al., 2008, 2009a). Hence, a higher degree of diffusion in ICT is to be expected in those territories with a higher proportion of urban population (Crenshaw & Robison, 2006; Liu & San, 2006). While research based on microdata has generally shown that individuals in urban locations and highly populated areas are more likely to use the Internet than those in rural and low populated areas (Chaudhuri et al., 2005; Mills & Whitacre, 2003; Noce & McKeown, 2008; US Department of Commerce, 2000), some cross-country studies find a non-significant or negative effect of urbanization and population density on ICT adoption (Chinn & Fairlie, 2007, 2010; Yartey, 2008).

Other authors have explored ICT demand from the approach of network effects. A technology is said to exhibit network effects or externalities when the value of the technology for a user increases with the total number of users in the network (Shapiro & Varian, 1999). In the particular case of the Internet, it is clear that the larger the number of people joining, the more communication possibilities and the more content online. Goolsbee and Klenow (2002) show evidence that households in regions with a high computer penetration are more prone to buy a first computer even once controlled for individual characteristics. Moreover, Whitacre and Mills (2007) confirm that network externalities, together with education and income, are the most important factors to explain the rural versus urban broadband. Schleife (2010) also finds evidence of the existence of such network effects and reports that, on average, a 10 percentage-point increase in the regional proportion of Internet experienced users raises the probability of being online by 1.8 percent. Furthermore, Andrés et al. (2010), using a panel of 214 countries during the period 1990–2004, identify that network effects are crucial in the worldwide diffusion of the Internet.

In addition, many researchers have highlighted the role played by institutional factors, commercial openness as well as cultural elements to explain the digital gap (Billón et al., 2010; Chinn & Fairlie, 2007, 2010; Corrocher & Ordanini, 2002; Estache, Manacorda, & Valletti, 2002; Hargittai, 1999; Norris, 2001; Sciadas, 2005; Van Dijk & Hacker, 2003).

The institutional legal environment in a territory is another key explanatory factor. In this sense, telecommunication policy has focused on the development of infrastructure, the liberalization of the market allowing for new entrants, and thus the reduction in ICT prices and costs. Hargittai (1999) is among the first to find evidence that those countries with a monopoly

in the telecommunication sector have significantly lower levels of Internet adoption. In the same line, [Bauer, Berne and Maitland \(2002\)](#) and [Guillén and Suárez \(2005\)](#) show that competition in local telephone service raises Internet penetration. More recently, [Andrés et al. \(2010\)](#) identify a positive correlation between the number of Internet service providers (ISP) and the rates of adoption. Nonetheless, a reduction in telecom prices might be necessary as well to really stimulate diffusion ([Kiiski & Pohjola, 2002](#)). It is interesting to note that [Wallsten \(2005\)](#) finds that regulation in this field leads to higher ISP final-user costs. However, the evidence regarding the role of prices is not yet clear ([Andonova, 2006](#); [Billón et al., 2010](#); [Chinn & Fairlie, 2007](#); [Hargittai, 1999](#)).

The quality of the institutions and government effectiveness are also associated with ICT penetration. Several authors have found rule of law as a significant factor to explain cross-country differences in Internet adoption ([Billón et al., 2009b](#); [Chinn & Fairlie, 2007, 2010](#); [Orviska & Hudson, 2009](#); [Zhao, Kim, Suh, & Du, 2007](#)). Good property rights protection ([Caselli & Coleman, 2001](#)), civil liberties ([Andonova, 2006](#); [Beilock & Dimitrova, 2003](#)), political stability ([Liu & San, 2006](#)), political rights, and investment climate also affect Internet diffusion ([Andonova, 2006](#)).

Likewise, the level of commercial openness of the economy may be relevant for ICT adoption. Although a positive and significant impact has been expected for this factor, the empirical evidence has thrown little light on such issue ([Caselli & Coleman, 2001](#); [Chinn & Fairlie, 2007, 2010](#); [Crenshaw & Robison, 2006](#); [Hargittai, 1999](#); [Kraemer et al., 2005](#); [Vicente & López, 2006b](#); [World Bank, 2009](#); [Yartey, 2008](#)).

A final set of variables, which have also been considered as potential determinants of ICT adoption, are those related to cultural factors. In this sense, special attention has been paid to the role of English. Since most online content is in English, individuals' language skills may influence their interest in the Net ([Hargittai, 1999](#)). Nevertheless, research has not produced any conclusive results ([Billón et al., 2009b](#); [Chinn & Fairlie, 2007](#); [Hargittai, 1999](#); [Kiiski & Pohjola, 2002](#); [Liu & San, 2006](#); [Tanner, 2003](#)). Evidence on other cultural influences is scarce due to data limitations. [Erumban and De Jong \(2006\)](#) find that cross-country differences in ICT adoption can be explained, at least to some extent, by power distance and uncertainty avoidance.

To sum up, it is worth noticing that this literature highlights the multidimensional nature of ICT and the digital divide: there are many ICT, many divides, and many socio-economic, regulatory, and cultural factors that might explain the digital disparities both within and between territories. Within this framework, next sections present the analysis of the regional digital divide which attempts, in the first place, to measure it, and then, to explain it.

3. Measuring the digital divide

As stated previously, the first goal of this paper is to measure the digital divide across European regions. The digital divide is a complex and multidimensional issue, which requires taking into account diverse technologies, variables, and territories ([Billón et al., 2009b](#); [OECD, 2009](#)). In this context, factor analysis has revealed as a useful tool to capture the overall dimension of the digital divide as shown by [Al-Mutawakil et al. \(2009\)](#), [Corrocher and Ordanini \(2002\)](#), [Soupizet \(2004\)](#), [Trkman, Blazic, and Turk \(2008\)](#), and [Vicente and López \(2006a, 2007\)](#), among others. Therefore, the regional digital divide is measured here by means of the application of this technique over a set of regional ICT indicators from Eurostat statistics.

3.1. Data

One of the main constraints to the analysis of the digital divide is associated with the availability of data on a regional basis. Despite the efforts devoted in this field by some European projects ([BISER, 2004](#); [UNDERSTAND, 2006](#)), there is still a lack of comparable regional ICT data. Hence, regional authorities frequently do not have enough information in order to properly track ICT adoption and evaluate the effects of ICT policies ([Billón et al., 2008](#)). As a consequence, there is usually a “trade-off” between the number of indicators and territories that can be included in the investigation ([Vicente & López, 2006a](#)).

Bearing in mind these limitations, Eurostat's regional information society statistics are used in order to measure the digital divide ([Eurostat, 2009a](#)). This data is based on Eurostat's annual model survey on ICT usage, which “aim is to collect and disseminate harmonized and comparable information on the use” of these technologies in households and by individuals in the EU ([Eurostat, 2009a](#)). Therefore, data is collected by the National Statistical Institutes or Ministries of each member state, usually through face-to-face or telephone interviews. The sampling universe consists of all the European population between 16 and 74 years old, and all the households with at least one member in this age range. Out of this universe, for each country an independent, representative, multiple-stratified random sample is drawn.

Both household and individual-level statistics are published. Household-level data is presented according to the number of adults and children in the household, as well as locality; while individual characteristics are disaggregated by age, gender, education level, employment situation, and location.

The provision of regional breakdowns by countries was voluntary for the years 2006 and 2007. In 2008, the collection of NUTS-1 breakdowns was set to be obligatory (NUTS-2 still being optional). That is the reason why the present study uses 2008 data, covering 164 regions at NUTS-1 and NUTS-2 levels (whenever possible) with the exceptions of Hungary, Romania, and Sweden, for which there is no regional disaggregated data. In spite of this, the three countries were included in the analysis in order to get a full picture of the EU. However, it is important to bear in mind such limitation when interpreting the results for those countries.

Table 1
Description of variables and codes.

CODE	Variable
HOME	Percentage of households with access to the Internet
BROAD	Percentage of households with a broadband connection
USE	Percentage of individuals regularly using the Internet, that is, those who accessed the Internet, on average, at least once a week
NOPC	Percentage of individuals who have never used a computer
COM	Percentage of individuals who ordered goods or services online for private use

Table 2
Descriptive statistics.
Source: Eurostat (2009a).

	HOME	BROAD	USE	NOPC	COM
Mean	58.39	47.27	56.13	27.68	29.84
Median	58.50	47.50	57.00	25.00	23.00
Std. Deviation	16.38	16.25	17.03	15.37	18.66
Skewness	−0.22	−0.08	−0.01	0.49	0.31
Kurtosis	−0.32	−0.72	−0.85	−0.84	−1.29
Minimum	15.00	12.00	22.00	5.00	1.00
Maximum	90.00	79.00	97.00	61.00	69.00
Percentile 25	47.00	35.00	44.00	14.00	14.00
Percentile 50	58.50	47.50	57.00	25.00	23.00
Percentile 75	71.00	58.75	70.00	39.00	48.75

Eurostat' survey on ICT usage by households and individuals focuses on the following five regional indicators (Table 1): percentage of households with access to the Internet (HOME); percentage of households with a broadband connection (BROAD); percentage of individuals regularly using the Internet, that is, those who accessed the Internet on average at least once a week (USE); percentage of individuals who have never used a computer (NOPC);⁴ and percentage of individuals who ordered goods or services over the Internet for private use (COM).

Eurostat does not publish any regional data on ICT adoption by businesses. However, it can be expected to be highly correlated with the general spread of these technologies across population: thus, the higher the rate of ICT adoption by households and individuals, the higher the use by firms as well.

Table 2 shows some descriptive statistics. They reflect several important differences across regions. For instance, the percentage of households connected to the Internet goes from a minimum of 15 percent in the region of Severozapaden (Bulgaria) to a maximum of 90 percent in Noord-Holland (the Netherlands). Moreover the percentage of individuals using e-commerce goes from 1 percent in Yugoiztochen (Bulgaria) to 69 percent in the South East of the United Kingdom.

3.2. Factor analysis

Factor analysis is a multivariate technique which allows transforming a set of variables into a group of new components through linear combinations of the original variables. The extraction method ranks the new components according to the share of explained variance (Hair, Anderson, Tatham, & Black, 1995).

First of all, it is necessary to assess the correlation between the variables in order to evaluate the appropriateness of this technique. The correlation matrix (Table 3) shows that all variables are closely related to each other, with correlation coefficients (in absolute values) larger than 0.8. For instance, the percentage of individuals who have never used a computer is highly and negatively correlated with using the Internet and having access at home because most people are connected to the Internet through a computer. Furthermore, the high and positive correlations of broadband with home access and use show the increasing importance of high-speed connections: many of the latest online services (such as e-commerce, e-learning or streaming video) are only possible through fast networks.

Given these strong correlations, factor analysis seems to be appropriate. Nonetheless, additional insights can be gained by means of both the Bartlett test of sphericity and the Kaiser–Mayer–Olkin (KMO) measure of sampling adequacy. The null hypothesis in the Bartlett test is that the correlation matrix is an identity one, or in other words, that there is no correlation between the variables. The KMO measure takes values between 0 and 1 (the higher the value the more suitable factor analysis

⁴ Eurostat considers the "percentage of individuals who have never used a computer" as an indicator of e-skills, and in particular, of the lack of digital skills across population. Nonetheless, it would be of more interest to have other indicators of digital skills, based on a self-assessment approach, such as those available at country level at Eurostat.

Table 3
Correlation matrix.

	HOME	BROAD	USE	NOPC	COM
HOME	1.00	0.92	0.94	–0.92	0.89
BROAD	0.92	1.00	0.93	–0.87	0.82
USE	0.94	0.93	1.00	–0.96	0.87
NOPC	–0.92	–0.87	–0.96	1.00	–0.90
COM	0.89	0.82	0.87	–0.90	1.00

Note: All the correlations are significant at the 1 percent level.

Table 4
Results of factor analysis.

Factor	Eigenvalue	Percent of variance	Cumulative percent of variance
1	4.60	92.04	92.04
2	0.20	3.91	95.95
3	0.11	2.22	98.17
4	0.06	1.27	99.44
5	0.03	0.56	100.00

Note: The extraction method is principal component analysis.

Table 5
Factor loadings matrix and communalities.

	Factor loadings	Communalities
USE	0.98	0.96
HOME	0.97	0.95
NOPC	–0.97	0.93
BROAD	0.95	0.90
COM	0.93	0.87

Note: The extraction method is principal component analysis.

is). In this case, the Bartlett statistic gets a value of 1386 with an associated probability of less than 1 percent, which suggests rejecting the null hypothesis, and the KMO measure is 0.85. All this indicates that the application of the technique is appropriate.

Therefore, factor analysis is run. Results are presented in Table 4. The initial solution shows as many factors as variables, that is, five; so the next step is to decide how many factors to extract. According to the eigenvalue criterion, all factors with eigenvalues larger than 1 should be retained in order to guarantee that each of them comprise at least the variance of a single variable. The percentage of variance criterion takes account of those factors explaining at least 60 percent (or even 80 percent) of the variance of the original variables. For the case of analysis, both criteria indicate that only the first factor should be considered. Such factor explains 92 percent of the total variance of the original variables.

Table 5 shows the factor loadings matrix, which includes the correlation coefficients between each factor and each variable, and the communalities. Factor 1 has significant positive loadings on all the variables except for the percentage of individuals who have never used a computer, which gets a negative sign. The variable with the highest loading is the percentage of individuals using the Internet, followed by home-Internet-access penetration. The communalities show that the retained factor explains between 87 and 96 percent of the variance of each individual variable. Such high values indicate the quality of the results.

By computing the factor score of each region, a ranking can be developed to identify best-and-worst performing territories. The main advantage of building such a ranking is that it provides a clear picture of regional performance. Furthermore, it is easier to interpret the place of a region on it than the position derived from a set of various separate indicators. However, such advantages can transform into disadvantages if simplistic conclusions are derived from it (OECD & JRC, 2008).

Table 6 shows that Nordic European regions are the most advanced when it comes to ICT. In particular, Dutch regions stand out with seven places among the Top-10, including the first position of Noord-Holland. These regions show extraordinary high levels of Internet penetration with about 85 percent of individuals regularly using the Internet, and almost 90 percent of households connected to the Net, mainly by high-speed connections. These high levels of access imply both high levels of usage (more than half of individuals buying online) and digital skills (less than 7 percent of population has never used a computer). The rest of the Top-10 corresponds to two Danish regions (ranks 3rd and 10th), and the British South East (rank 8th).

Table 6

Ranking of regions derived from factor analysis.

Rank	Region and code	Score	Rank	Region and code	Score	Rank	Region and code	Score
1	nl32 Noord-Holland	1.82	32	deb Rheinland-Pfalz	1.02	63	fr7 Center-Est	0.38
2	nl31 Utrecht	1.80	33	ukm Scotland	0.98	64	at11 Burgenland	0.38
3	dk01 Hovedstaden	1.78	34	de6 Hamburg	0.95	65	fr5 Ouest	0.37
4	nl11 Groningen	1.78	35	dec Saarland	0.95	66	dee Sachsen-Anhalt	0.36
5	nl21 Overijssel	1.68	36	de1 Baden-Württemberg	0.95	67	ukc North East	0.28
6	nl33 Zuid-Holland	1.62	37	ukf East Midlands	0.95	68	be10 Région de Bruxelles-Capitale	0.28
7	nl23 Flevoland	1.61	38	at13 Wien	0.94	69	de8 Mecklenburg-Vorpommern	0.25
8	ukj South East	1.58	39	de9 Niedersachsen	0.93	70	fr4 Est	0.24
9	nl22 Gelderland	1.58	40	de5 Bremen	0.92	71	es30 Comunidad de Madrid	0.23
10	dk04 Midtjylland	1.54	41	def Schleswig-Holstein	0.90	72	fr2 Bassin Parisien	0.23
11	se Sweden	1.51	42	ukl Wales	0.89	73	at21 Kärnten	0.22
12	nl12 Friesland	1.41	43	fi13 Itä-Suomi	0.87	74	de4 Brandenburg	0.22
13	dk02 Sjælland	1.39	44	uke Yorkshire and The Humber	0.85	75	fr3 Nord-pas-de-Calais	0.21
14	nl41 Noord-Brabant	1.39	45	ukg West Midlands	0.82	76	ded Sachsen	0.20
15	fi20 Åland	1.33	46	be31 Prov. Brabant Wallon	0.77	77	be25 Prov. West-Vlaanderen	0.13
16	dk03 Syddanmark	1.32	47	be24 Prov. Vlaams Brabant	0.76	78	fr6 Sud-Ouest	0.12
17	uki London	1.29	48	fr1 Île de France	0.75	79	ie Ireland	0.12
18	nl42 Limburg	1.27	49	at32 Salzburg	0.61	80	sk01 Bratislavský kraj	0.08
19	fi18 Etelä-Suomi	1.27	50	be23 Prov. Oost-Vlaanderen	0.60	81	es51 Cataluña	0.07
20	fi19 Länsi-Suomi	1.24	51	be22 Prov. Limburg	0.60	82	be34 Prov. Luxembourg	0.06
21	nl34 Zeeland	1.24	52	be21 Prov. Antwerpen	0.59	83	be35 Prov. Namur	0.05
22	ukh Eastern	1.24	53	at34 Vorarlberg	0.58	84	sk02 Západné Slovensko	0.03
23	ukk South West	1.19	54	deg Thüringen	0.57	85	be33 Prov. Liège	-0.02
24	dk05 Nordjylland	1.15	55	ukd North West	0.54	86	sk03 Stredné Slovensko	-0.03
25	lu Luxembourg (Grand-Duché)	1.12	56	at33 Tirol	0.52	87	ee Estonia	-0.04
26	nl13 Drenthe	1.11	57	at31 Oberösterreich	0.50	88	sk04 Východné Slovensko	-0.09
27	de3 Berlin	1.11	58	cz01 Praha	0.49	89	es53 Illes Balears	-0.10
28	de2 Bayern	1.10	59	fr8 Méditerranée	0.48	90	si Slovenia	-0.18
29	dea Nordrhein-Westfalen	1.07	60	at22 Steiermark	0.42	91	es21 Pais Vasco	-0.19
30	de7 Hessen	1.05	61	at12 Niederösterreich	0.40	92	cz02 Střední Čechy	-0.21
31	fi1a Pohjois-Suomi	1.04	62	ukn Northern Ireland	0.40	93	es22 Comunidad Foral de Navarra	-0.24
94	es13 Cantabria	-0.27	118	pl2 Poludniowy	-0.66	142	ite3 Marche	-1.08
95	be32 Prov. Hainaut	-0.29	119	itd5 Emilia-Romagna	-0.68	143	cy Cyprus	-1.12
96	es24 Aragón	-0.32	120	ite4 Lazio	-0.69	144	itc3 Liguria	-1.13
97	es23 La Rioja	-0.33	121	cz07 Střední Morava	-0.70	145	pt11 Norte	-1.20
98	mt Malta	-0.34	122	es41 Castilla y León	-0.73	146	itf1 Abruzzo	-1.22
99	lv Latvia	-0.35	123	itd2 Provincia Autonoma Trento	-0.75	147	pt20 Região Autónoma dos Açores	-1.25
100	pt17 Lisboa	-0.35	124	gr3 Attiki	-0.75	148	pt18 Alentejo	-1.29
101	es12 Principado de Asturias	-0.36	125	es61 Andalucía	-0.75	149	pt16 Centro	-1.30
102	hu Hungary	-0.42	126	cz04 Severozápad	-0.79	150	itf2 Molise	-1.31
103	es70 Canarias	-0.43	127	itd4 Friuli-Venezia Giulia	-0.81	151	itf5 Basilicata	-1.31
104	cz03 Jihozápad	-0.52	128	es42 Castilla-la Mancha	-0.82	152	itf3 Campania	-1.48
105	itd1 Provincia Autonoma Bolzano-Bozen	-0.53	129	pt15 Algarve	-0.84	153	itf6 Calabria	-1.55
106	cz06 Jihovýchod	-0.54	130	es62 Región de Murcia	-0.87	154	itf4 Puglia	-1.60
107	es52 Comunidad Valenciana	-0.54	131	es43 Extremadura	-0.91	155	itg1 Sicilia	-1.62
108	es64 Ciudad Autónoma de Melilla	-0.56	132	itd3 Veneto	-0.93	156	gr4 Nisia Aigaiou. Kriti	-1.87
109	pl1 Centralny	-0.58	133	itc2 Valle d'Aosta/Vallée d'Aoste	-0.94	157	ro Romania	-1.87
110	pl5 Poludniowo-Zachodni	-0.58	134	pl3 Wschodni	-0.94	158	gr1 Voreia Ellada	-1.88
111	pl4 Północno-Zachodni	-0.58	135	ite2 Umbria	-0.97	159	bg13 Severoiztochen	-1.89
112	itc4 Lombardia	-0.64	136	es11 Galicia	-0.99	160	bg22 Yuzhen tsentralen	-1.94
113	es63 Ciudad Autónoma de Ceuta	-0.64	137	ite1 Toscana	-1.01	161	bg12 Severen tsentralen	-1.96
114	cz05 Severovýchod	-0.64	138	itc1 Piemonte	-1.03	162	bg23 Yugoiztochen	-1.99
115	lt Lithuania	-0.65	139	bg21 Yugozapaden	-1.04	163	bg11 Severozapaden	-2.09
116	pl6 Północny	-0.66	140	itg2 Sardegna	-1.05	164	gr2 Kentriki Ellada	-2.09
117	cz08 Moravskoslezsko	-0.66	141	pt30 Região Autónoma da Madeira	-1.07			

Note: See Appendix for countries' abbreviations.

On the opposite, the Greek region of Kentriki Ellada is in the last position of the ranking (rank 164th), preceded by five Bulgarian regions, two other Greek and one Italian regions. It is interesting to note that Bulgaria places all its regions among the Bottom-10 with the exception of the one which corresponds to the capital city, ranking 20 places ahead (Yugoizapaden in the 139th position).

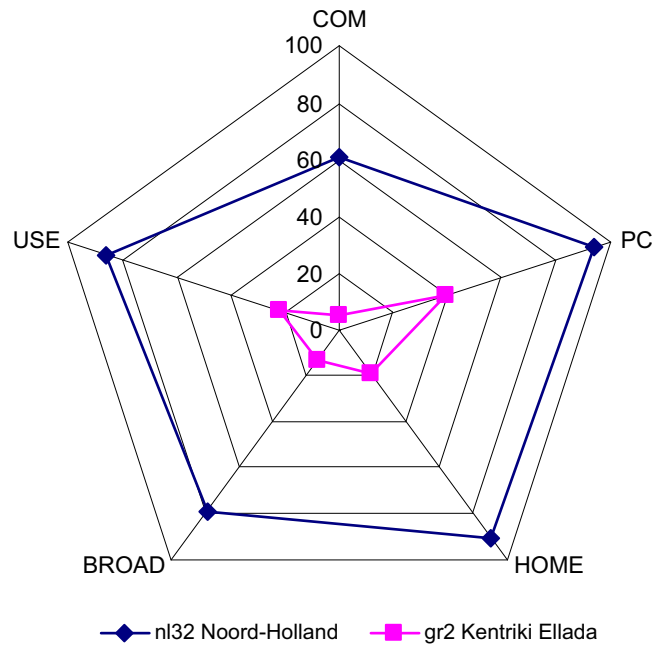


Fig. 1. The digital divides between top and bottom regions.

Romania is also positioned at the Bottom-10, but it is important to recall that there is no disaggregated information for this country: therefore, its 157th rank could be disguising serious regional digital disparities within it. The same could be happening with Sweden (rank 11th) and Hungary (rank 102nd) which, despite the lack of disaggregated data, were included in the analysis in order to have a full picture of the EU-27.⁵

Fig. 1 shows a diamond chart for the top and bottom regions, that is, Noord-Holland (1st) versus Kentriki-Ellada (164th). In order to give a clear picture of the magnitude of the regional gap across the EU-27, the variable “percentage of individuals who have ever used a computer” has been included, instead of the “percentage of those who have never used it” (NOPC).⁶ The chart shows that the digital divides between the top and bottom regions are big indeed. In fact, the differences in all the indicators exceed 50 percentage points. Moreover, the biggest gap is observed for home-Internet-connection with 71 percentage points of difference between Noord-Holland (90 percent) and Kentriki-Ellada (19 percent).

Looking further into the ranking it is also observed that the first half of it mostly corresponds to regions from Northern and Central European countries. The only exceptions are Prague and Madrid ranking 58th and 71st, respectively. From the place 86th on, all the regions belong to Southern countries and the new members of the Union, with the only exception of the Belgian region of Hainaut which ranks 95th.

The fairly good place of Prague, ahead of some Northern regions, contrasts with the position of the rest of Czech territories which rank between 92nd and 126th places. The same happens with other countries such as Belgium, whose regions rank between 46th and 95th positions. These results suggest the existence of important divides within countries. Further insights on such issue are presented in Fig. 2 which shows for each country the best-and-worst performing regions, and the national average. The horizontal axis identifies countries by their abbreviations while the vertical axis displays factor scores with the value of zero corresponding to the European average. Hence regions in the upper part of the diagram are those over the average in what regards to ICT. If the domestic digital divide is measured as the difference between the best and the worst performing regions in each country, Greece and the United Kingdom present the biggest divides. However, the regions of the former are all below the European average, while those of the latter are all above it. The Czech Republic and Spain also have big digital gaps with the peculiarity that best performing regions are above the European average in contrast to the worst ones that are below it. The same happens in Belgium. Overall the chart points out the leading position of Northern regions in contrast to the Southern and Eastern ones.

Before getting further deep into the analysis of the factors that might explain the observed disparities, some comparisons have been made between the factor-derived ranking and some other rankings resulting from several ICT indices.⁷ Since there are no previous measurements of the digital divide across European regions to compare with, it was necessary to take into

⁵ The results for these three countries are in line with those of the mid-term review of the plan i2010 (European Commission, 2008b). Sweden was found to be positioned in the most-advanced group of countries with more than 75 percent of ICT indicators over the EU average, in contrast to Hungary and Romania which were below that average.

⁶ The percentage of individuals who have ever used a computer has been reckoned as the following difference = 100 – NOPC (percentage of individuals who have never used a computer).

⁷ This point was suggested by one of the reviewers.

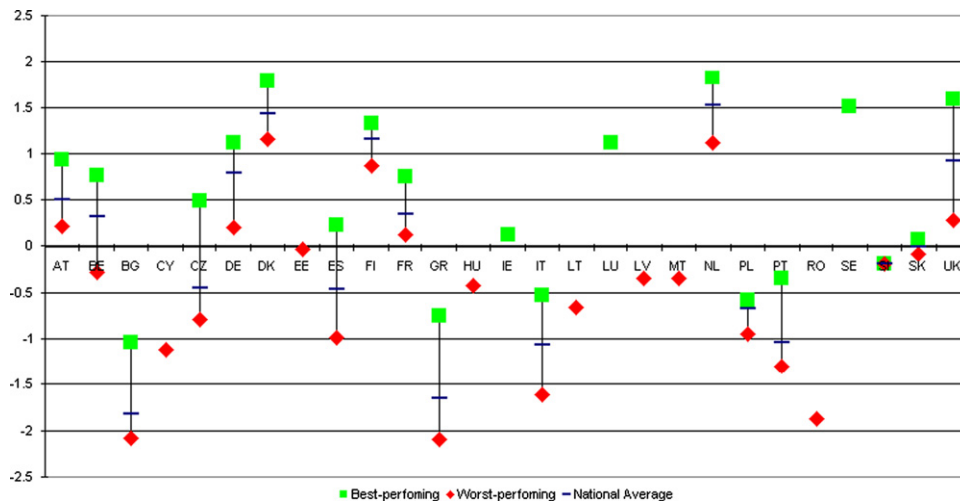


Fig. 2. The digital divides within countries. Note: See Appendix for countries' abbreviations.

Table 7

Rank correlation matrix (Spearman's rank correlation coefficient).

Sources: (1) It is the national version of the index derived from factor analysis; (2) 2008 network readiness index rankings by WEF (2008); (3) 2007 ICT development index rankings from ITU (2009); (4) 2008 European broadband performance rankings from the European Commission (2008a); (5) 2008 e-readiness rankings from the Economist Intelligence Unit (2009).

	FA (1)	WEF (2)	ITU (3)	EU (4)	EUI (5)
FA	1	0.88	0.85	0.84	0.84
WEF	0.88	1	0.90	0.85	0.93
ITU	0.85	0.90	1	0.83	0.95
EU	0.84	0.85	0.83	1	0.90
EUI	0.84	0.93	0.95	0.9	1

Notes: The EU's 27 member states are considered, except for the correlations with the EUI's e-readiness ranking for which only information on 25 countries is available. All the correlations are significant at the 1 percent level.

account some international and European-level indices; in particular, the following have been considered: the 2008 Network Readiness Index by the WEF (2008); the 2007 ICT Development Index from the ITU (2009); the 2008 European Broadband Performance Index from the European Commission (2008a); and the 2008 e-Readiness Index from the Economist Intelligence Unit (2009).⁸ The comparison with such indices required to recalculate a national version of the regional factor-derived index.

When comparing rankings, the first point to examine is the extent to which they are similar (Archibugi & Coco, 2005). For this purpose, Spearman's rank correlation coefficient is used (Table 7). Results show a high correlation (over 0.8) between the rankings of each pair of indices, with the highest found between ITU and EUI (0.95). In the case of the factor-derived ranking, the correlations keep below the threshold of the 0.9 level, finding the highest with WEF (0.88). It is worth noticing that value, since the WEF ranking is derived from a set of 68 variables, grouped into three categories (environment, readiness, and usage), compared to the five regional indicators employed in the present analysis, and the eleven variables considered by the ITU. Such result might indicate, at least to some extent, that working with detailed and disaggregated data can provide a full picture of a country's digital position as good as the one obtained using a high number of country-level variables.

The second point to evaluate refers to changes in the rank of countries depending on the index used (Archibugi & Coco, 2005). In Table 8, columns 1–5 show the rankings and columns 6, 7 and 8 indicate the rank mean, the rank median, and the standard deviation, respectively. Since the rank-correlations between the indices are very high as previously observed, few important differences can be noticed. In fact, the biggest discrepancy between the five indices refers to the position of Italy. In the factor-derived ranking, Italy is the 23rd; a bit better for the WEF, the 21st; and moving up in the rest of the indices, reaching the 10th position according to ITU. The poor Italian performance in the factor analysis can be mainly explained by the fact that more than a half of its regions are at the Bottom-15, and the other half are fairly below the European average.

⁸ Despite the fact that this index does not include information for the full EU (Cyprus and Luxembourg are missing), it has been taken into account since it is one of the most relevant and consolidated measures in this field, being annually published since the year 2000.

Table 8

Rankings of European countries according to several ICT indices.

Sources: See notes under Table 7.

	FA (1)	WEF (2)	ITU (3)	EU (4)	EUI (5)	Rank mean	Rank median	Standard deviation
NL	1	4	3	2	3	2.5	2.5	1.12
SE	2	2	1	1	1	1.5	1.5	0.50
DK	3	1	2	3	2	2.25	2.5	0.83
FI	4	3	5	8	6	5	4.5	1.87
LU	5	11	4	11	n.a.	7.75	8	3.27
UK	6	5	6	4	4	5.25	5.5	0.83
DE	7	7	7	9	7	7.5	7	0.87
AT	8	6	9	7	5	7.5	7.5	1.12
FR	9	9	11	5	10	8.5	9	2.18
BE	10	12	12	6	8	10	11	2.45
IE	11	10	8	19	9	12	10.5	4.18
SK	12	22	23	21	20	19.5	21.5	4.39
EE	13	8	13	20	15	13.5	13	4.27
SI	14	15	15	12	16	14	14.5	1.22
MT	15	13	16	13	11	14.25	14	1.30
LV	16	23	21	18	22	19.5	19.5	2.69
HU	17	19	20	17	19	18.25	18	1.30
CZ	18	18	25	15	18	19	18	3.67
ES	19	16	14	10	13	14.75	15	3.27
LT	20	17	18	22	21	19.25	19	1.92
PL	21	26	24	24	23	23.75	24	1.79
PT	22	14	17	14	14	16.75	15.5	3.27
IT	23	21	10	16	12	17.5	18.5	5.02
CY	24	20	22	26	n.a.	23	23	2.24
GR	25	24	19	23	17	22.75	23.5	2.28
BG	26	27	26	27	25	26.5	26.5	0.50
RO	27	25	27	25	24	26	26	1.00

Notes: 'n.a.' stands for not available.

The rank mean and median have been reckoned considering the ranks from the first four indices because the fifth index has not complete information for the full EU (CY and LU are missing). The same happens with the standard deviation.

All the considered indices usually refer to a bigger sample of countries than the EU. Since the analysis focuses on the latter area, the rankings have been re-scaled to a range from 1 to 27.

4. Explaining the digital divide

4.1. Explanatory factors

Once measured the European regional digital divide, the second goal of this paper is to explain the observed disparities. The choice of the explanatory variables is based on the specialized literature as exposed in Section 2; however it should be noticed that this selection has been restricted by data availability. As a consequence, some potentially important determinants (such as Internet access costs) could not be included due to the lack of disaggregated information at regional level.

In this context, the first explanatory factor to consider is regional economic development measured by Gross Domestic Product (GDP) per capita. The variable unemployment rate is also included since economic development is clearly related to the situation of the labor markets (Billón et al., 2008, 2009a; Schleife, 2010).

Taking into account the above mentioned impact of human capital on ICT adoption (Demoussis & Giannakopoulos, 2006; Gibson, 2003; Pohjola, 2003; Quibria et al., 2003), the two following variables are considered: the percentage of population aged 15 and over who have completed tertiary education, as previously done by Billón et al. (2008, 2009a); and the percentage of human resources in science and technology activities over total population, which is a proxy for regional availability of skilled workforce.

The link between ICT and regional innovation performance is also examined. The economic literature on technological diffusion emphasizes the role of innovative activities in facilitating the successful use of external knowledge in general and of new technologies in particular (Cohen & Levinthal, 1989; Karshenas & Stoneman, 1995). Hence, best-performing territories in innovation will be better positioned to adopt ICT and, therefore, to take advantage of the benefits that these technologies may provide (Maurseth & Frank, 2009). In this sense, Norris (2001) and Vicente and López (2006b) found, for a set of 179 countries and the European Union, respectively, that those nations which spent more on research and development were more prone to adopt ICT. In order to take account of innovation performance across European regions, the Regional Innovation Score (RIS) has been considered. The RIS was developed at the proposal of the European Commission "to evaluate and compare the innovation performance" of the member states under the Lisbon Strategy (Hollanders, 2007).

Since the EU has been paying special attention to bridging the rural–urban digital divide (Dabinett, 2001; European Commission, 2006; Technopolis & IRISI-Europe, 2002), the relationship between ICT factor scores and population density has

Table 9

Correlation matrix (Pearson correlation coefficient).

Source: Eurostat (2009) and [Hollanders \(2007\)](#).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) ICT factor score	1.00								
(2) GDP per capita	0.67	1.00							
(3) Unemployment rate	−0.30	−0.24	1.00						
(4) Population with tertiary studies (percentage of population)	0.53	0.45	0.08	1.00					
(5) HH.RR. in Science & Technology (percentage over total population)	0.76	0.62	−0.24	0.76	1.00				
(6) Regional Innovation Score (RIS)	0.73	0.62	−0.39	0.59	0.77	1.00			
(7) Population density (inhabitants per square kilometer)	0.17	0.35	0.29	0.32	0.31	0.31	1.00		
(8) Population > 65 years old (percentage of population)	−0.18	0.02	0.07	−0.13	−0.01	−0.02	−0.16	1.00	
(9) Employment in services (percentage over total employment)	0.47	0.62	0.07	0.39	0.47	0.45	0.36	0.06	1.00

Note: GDP and HH.RR. stand for Gross Domestic Product and Human Resources, respectively. The number of observations is 164, except for the correlations with RIS (151) and HH.RR. in Science and Technology (160). The correlation between RIS and HH.RR. is based on 147 observations.

also been analyzed. Furthermore, the percentage of population aged 65 and over is included to capture the potential influence of regional demographic differences on the extent of ICT adoption. In this sense, older people often face more barriers to use ICT than younger individuals ([Cohendet, 2003](#); [Holloway, 2005](#)).

The connection between regional specialization and ICT is explored as well, given previous evidence on the impact on industry composition over territorial inequalities across the EU ([European Commission, 1999](#); [Neven & Gouyette, 1995](#)). Some prior research has found that the spatial pattern of Internet diffusion can be explained at least to some extent by the differences in productive structures between European regions ([Billón et al., 2008, 2009a](#)). Since services are intensive users of ICT ([Horrigan et al., 2006](#)), a high share of this industry in the regional economy is expected to be positively correlated with ICT adoption. Therefore, the link between the share of regional employment in services and ICT is considered.

Finally, some cultural and institutional factors for regional ICT adoption are taken into account. Several authors point out that when a small group of territories is to be analyzed, minor changes in policy and institutional actions appear to have major impacts over technology adoption ([Guillén & Suárez, 2001](#); [Howard, Anderson, Busch, & Nafus, 2009](#); [Oyeleran-Oyeyinka & Lal, 2005](#)). Since it is very difficult to find data on such issues available at the regional level, the following two dummy variables are used as proxies: the first one is related to whether the population in a region is native-English speaker (that would be the case of Irish and British territories), and the second one refers to whether a region belongs to a Nordic country (Denmark, Finland, and Sweden).⁹ The reason to include the latter variable is that these countries were among the first to implement strategies to foster the development of the information society.

The data on the set of explanatory variables comes from [Eurostat \(2009b\)](#), with the exception of the information about the Regional Innovation Score (RIS) which is drawn from [Hollanders \(2007\)](#).

4.2. Correlation analysis

[Table 9](#) shows the correlations between the derived ICT scores from factor analysis and the quantitative socio-economic indicators mentioned above.¹⁰ The highest correlations with ICT scores are found for GDP per capita, human resources in science and technology, and RIS, in contrast to population density, which shows the lowest among the considered variables.

Results here indicate that the relationship between the rates of ICT adoption and the levels of income is a bit less strong at the regional level than those reported at country level: a value of almost 0.7 is reckoned compared to the 0.9 found by some authors at country level ([Hanafizadeh, Saghaei, & Hanafizadeh, 2009](#); [Orbicom, 2005](#)). Getting a bit more detail, it is worth noticing that some regions such as Luxemburg and Bruxelles are not doing as well in ICT as they should be with respect to their level of income; while others such as Flevoland and Overijssel are doing much better in ICT than their incomes would suggest. Although these results must be taken with caution because of the time lag between the available data (the years 2006 and 2008 for GDP and ICT, respectively), they could be interpreted as an indication that other factors, apart from income, come into play to explain ICT adoption. One of these factors could be regional policy. This might be the case of the aforementioned Dutch regions of Flevoland and Overijssel. Both of them, being Objective 1 and 2 areas, respectively with incomes below the European average,¹¹ received special funding for enabling ICT use and infrastructure from the 2000–2006 Cohesion Policy Programmes ([Applica, Ismeri Europa, & Wiiw Consortium, 2008](#)). Such funding could be the reason why these regions are performing in ICT better than the expected from their income levels.

As happens with income, human capital and innovation measures are positively correlated with ICT scores. It is worth noticing the lower correlation of the variable percentage of population with tertiary studies (0.5) compared to those of human

⁹ Some other dummies were initially considered, such as whether a region was Objective 1 or whether it belonged to a new member state, but in the end this option was rejected due to their high correlations with income.

¹⁰ The dummy nature of the variables related to English and being a Nordic region prevents from reckoning the correlations with the other quantitative variables.

¹¹ During the period 2000–2006 the province of Flevoland was a (phasing out) Objective 1 region.

resources in science and technology and RIS (over 0.7). Therefore, it seems that regional position in ICT is more correlated to innovation capacity and to the availability of skilled workforce than to having a big share of population with the highest educational attainment. However, the variables related to innovation and skilled workforce are not able to fully grasp the regional differences in ICT. In fact, the Top-10 innovation performing regions according to the 2006 RIS¹² do not coincide with those identified in the factor-derived ranking. Nonetheless, this comparison of the Top-10 has to be taken with caution since regional data on ICT adoption is not available for Sweden.

The correlation between ICT factor scores and population density might reveal some insights of the rural–urban digital gap. In this case, the coefficient gets a value of 0.17, indicating a very weak connection between these two variables. Therefore, it seems that the rural–urban digital gap is not that much. Although this result could be a sign of the success of the aforementioned institutional efforts towards bringing the rural gap, further exploration would be needed to draw robust conclusions about this fact.

Concerning the rest of the variables (unemployment, older population, services share in employment), their correlation coefficients (less than 0.5) indicate the lack of a strong relationship with ICT scores.

4.3. Regression model

Since all these factors individually considered explain the digital divide just partially, an econometric model is proposed in order to determine their joint role in explaining ICT adoption. Thus, the model can be expressed in matrix notation as the following:

$$y = X\beta + \xi \quad (1)$$

where y is a vector of the ICT factor scores obtained for each of the analyzed regions, X is a matrix including the regional values of the explanatory variables already mentioned, and vector ξ is the corresponding disturbance term. Finally, β is the vector of coefficients to be estimated.

Given that the dependent variable is a synthetic index (the scores obtained by factor analysis), the regression model explains just the information common to the original set of variables. In the present case, this is not a concern because almost all the total initial variability is captured by the obtained factor. As shown in Table 4, the first factor explains 92 percent of the total variance of the original variables.

Two other relevant issues must be considered with regard to the estimation of Eq. (1). In the first place, given that some of the explanatory variables may be contemporaneously correlated with the error term and then cause endogeneity problems, their lagged values are used in the estimation (Billón et al., 2008, 2009a; Rodríguez-Pose & Vilalta-Bufí, 2005). Depending on data availability, 2006 or 2007 values are employed. In the second place, the high correlations between the three variables related to human capital and innovation could lead to multicollinearity problems if all considered in the model.¹³ For this reason, only the variable with the highest correlation with ICT scores is incorporated in the final estimation, that is, human resources in science and technology activities as a percentage of total population.

Table 10 shows the results of the estimation. They confirm previous evidence on the positive impact of income level and human capital on ICT adoption. Hence, the higher the income, the more resources that can be invested in new technologies and therefore the higher the ICT score. Likewise, a large regional proportion of high-skilled workforce leads to a high rate of ICT adoption. As these workers improve their digital abilities at work, they tend to use computers and the Internet at home more often than other type of employees (Schleife, 2010). In contrast, unemployment rate and having an old population exert a negative influence. Although unemployed people could take advantage of using the Internet for job search and training activities, results show that above all unemployment imposes financial restrictions that constrict ICT adoption. In the same way, a high proportion of old population might imply a low level of abilities and skills to deal with new technologies and, consequently, a low rate of adoption.

No significant evidence is found for population density, a result which reinforces the idea that the rural versus urban digital divide is not that much, at least in the year 2008. Neither is significant the share of services in employment. These results contrast with those obtained by Billón et al. (2008, 2009a) who find these two variables to be significant. Such discrepancy may be related to the time period. Their data refers to the year 2002, an early period in which the Internet was still in the first stages of diffusion in the EU. In this context, the first ones to adopt were those individuals who had an easy and cheap access to the Internet because of living in urban areas. Likewise, regions with a high specialization in services soon took up the Internet and ICT, since these activities were the first to realize the benefits ICT could provide for them. The 2008 picture is completely different: the Internet has spread widely over population and over the whole economy.

The two dummy variables included in the estimation are significant and with positive signs, indicating that both Nordic and native-English speaker regions have some kind of advantage in ICT adoption over the rest of European territories. However, the significance of the English variable could be a signal, at least to some extent, of other underlying factors which favor British and Irish regions, such as the existence of important ICT clusters on the supply side (Quah, 2001).

¹² According to the 2006 RIS, the Top-10 includes the following regions: Stockholm (SE), Västsverige (SE), Oberbayern (DE), Etelä-Suomi (FI), Karlsruhe (DE), Stuttgart (DE), Braunschweig (DE), Sydsvrige (SE), Île de France (FR), and Östra Mellansverige (SE).

¹³ These high correlations are due to the fact that the RIS is built, among other variables, upon the two considered, population with tertiary education and human resources in science and technology.

Table 10
Regression results.

Dependent variable: ICT factor scores	Coeff.	Robust std. errors	<i>t</i>	<i>p</i>
GDP per capita	0.005	0.002	2.81	0.006
Unemployment rate	−0.027	0.014	−1.92	0.057
Human Resources in Science & Technology	0.066	0.008	8.64	0.000
Population density	0.014	0.051	0.26	0.792
Population over 65 years old	−0.670	0.315	−2.13	0.035
Employment in services	0.457	0.396	1.15	0.250
Native English Speaking (IE, UK)	0.486	0.179	2.72	0.007
Nordic regions (FI, DK, SE)	0.595	0.185	3.21	0.002
Constant		0.607	−6.14	0.000
<i>N</i>	162			
<i>F</i> -statistic	44.57			0.000
<i>R</i> -squared	0.7036			

Note: See Appendix for countries' abbreviations.

Although the value of the *R*-squared is fairly high (0.7), there is still a 30 percent of the total variation in regional ICT scores which remains unexplained. This result points out that there are still important differences across regions which are very likely to influence ICT take-up. Nevertheless, it might be quite difficult to model these factors due possibly to their non-economic nature as it would be the case, for instance, of the impact of the institutional framework and cultural factors (Beilock & Dimitrova, 2003). Furthermore, it is important to recall that the lack of regional data prevented from considering other elements that may play an important role in explaining regional digital disparities.

Finally, the results of the Breusch-Pagan tests indicate no signs of heteroskedasticity. Moreover, the variance inflation factor gets an average value of 1.65, which is within the acceptable limits defined by Belsley, Kuhn, & Welsh (1980) and indicates that the degree of collinearity among the explanatory variables is reasonably moderate.

5. Conclusion remarks

This paper has presented an analysis of the digital divide combining the two main approaches in the literature, measurement and explanation of the digital gap, in contrast to previous research which has primarily focused on one of these issues. In particular, the regional digital divide is measured not only across the full EU but also within each country. The sample consists of 164 regions covering the EU-27, which extends the area considered in previous studies. In addition, this paper distinguishes from prior research in the fact that cultural and institutional influences are examined as determinant factors of regional ICT adoption.

Factor analysis is employed in order to measure the gap and deal with its complexity. Using a set on five regional ICT-indicators, the application of this technique has led to the identification of just one dimension which summarizes the different aspects of regional ICT adoption. By computing scores on the identified factor, a ranking of regions is developed.

Results show that by the year 2008 the digital divide across European regions is still huge despite the EUR 5.5 billion spent on the information society by the 2000–2006 Structural Funds.¹⁴ Although such actions might have been successful in increasing penetration levels across the Union, regional disparities are big indeed. Hence, the gap between the identified top and bottom regions (Noord-Holland and Kentriki-Ellada, respectively) exceeds 50 percentage points in all ICT indicators. Furthermore, such financial efforts have not been able to make the regional digital divide stop reflecting the regional wealth gap. The ICT ranking clearly shows the leading position of Northern regions in contrast to Southern and Eastern territories. Thus, the Top-10 includes British, Danish, and Dutch regions compared to the Bottom-10 which consists of Bulgarian, Greek, and Italian regions.

In addition, the existence of important divides within countries can be highlighted. If the domestic digital divide is measured as the difference between the best and the worst performing regions in each country, Greece and the United Kingdom present the biggest gaps. However, the digital profile of these countries is completely different. Hence, regions of the former are all below the European average, while those of the latter are all above it. The Czech Republic and Spain also have big digital gaps with some regions above the European average and others below it.

Another interesting outcome is that results confirm those of several ICT indices which are based on a higher number of country-level variables than that considered at regional level. This could be interpreted as an indication of the fact that working with disaggregated data can provide a full picture of a country's digital position as good as the one obtained using a high number of aggregate variables.

In order to throw some more light over the regional digital divide, a causal analysis of the observed disparities is run by means of the estimation of a regression model in which the dependent variable is the obtained regional factor score. The independent variables are based on the literature on ICT adoption, with the peculiarity that two dummy variables are

¹⁴ Authors' calculations based on data from Sweco (2008).

incorporated as proxies for cultural and institutional influences. Although the role of this kind of factors has been previously studied at country level, this is the first time to analyze it at regional level.

The estimates show that regions' score on ICT has a positive and significant correlation with the level of GDP per capita and the availability of a skilled workforce measured by the percentage of human resources in science and technology over total population. Moreover, results point out that Nordic regions and those being native-English speakers have some advantage in ICT over the rest. In contrast, regional unemployment rate and the percentage of population over 65 years old are negatively correlated with scores. No significant impacts are found for either the share of services employment in the economy and population density. The latter result implies that the regional digital divide is not an urban–rural gap. Such outcome could be pointing at the success of recent actions towards bridging such dimension of the digital divide, though further research would be needed in this area.

Overall, results indicate that there is still a lot to do for European policies in order to reduce territorial imbalances in ICT adoption and use. It is crucial that actions are implemented to bridge not only the gap across the full Union but also the disparities within each member state. Special attention should be paid to the cases of Greece, the United Kingdom, the Czech Republic and Spain, where the biggest internal divides have been detected. The British case might not seem to be a priority, since its regions are all above EU average. However, the persistence of such internal gaps might lead to a serious increase in social and economic inequality as ICT are more and more important for the competitiveness of individuals, firms, and territories. Moreover, findings suggest that additional efforts should be devoted to improving digital skills of both the workforce and the eldest, as well as to developing the English skills of Southern and Eastern Europe populations.

Nonetheless, some limitations must be considered in the analysis, mainly given by data constraints. With regard to the measurement of the digital divide, only five variables are included. Hence, some of its aspects may not be covered. Furthermore, the regional digital divide is assessed at a given point of time. Nevertheless, a complete analysis of the gap requires to study its evolution. Besides, the time lag between the available data for ICT and GDP prevents from analyzing the role that ICT are playing in regional economic growth. In a recent paper (Forman, Goldfarb, & Greenstein, 2009) found that the use of ICT was only connected with wage growth in those American counties that were already wealthy in terms of income, education, population, and industry. Regarding the explanation of the gap, the lack of data has led to consider cultural and institutional influences as two dummy variables. Nevertheless, a deep comprehension of the role of these factors would require more detailed data and variables.

As new waves of regional data become available, future research should focus in analyzing both whether the identified digital gaps are narrowing or widening, and the role played by social, cultural, and institutional factors in explaining them. Moreover, another key matter is the analysis of the wireless digital gap at the regional level, since the access through these technologies is becoming more and more important. Likewise, it is crucial to measure the quality of connections in order to assess to what extent regions are ready for the next generation of online services and applications.

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Appendix

See Table 11.

Table 11
List of abbreviations.

Abbreviation	Country	Abbreviation	Country
AT	Austria	LV	Latvia
BE	Belgium	LT	Lithuania
BG	Bulgaria	LU	Luxembourg
CZ	The Czech Republic	MT	Malta
CY	Cyprus	NL	The Netherlands
DK	Denmark	PL	Poland
EE	Estonia	PT	Portugal
FI	Finland	RO	Romania
FR	France	SI	Slovenia
DE	Germany	ES	Spain
GR	Greece	SE	Sweden
HU	Hungary	SK	The Slovak Republic
IE	Ireland	UK	The United Kingdom
IT	Italy		

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